Case Study 17: Student Performance Trend Analysis

Complete Activity-Based Learning Report

1. Problem Statement

Compare student marks across multiple examinations, calculate improvements and declines, identify weak subjects, and classify student performance.

2. Objectives

• Compare examination marks.
• Calculate improvement or decline.
• Identify weak subjects.
• Classify student performance.

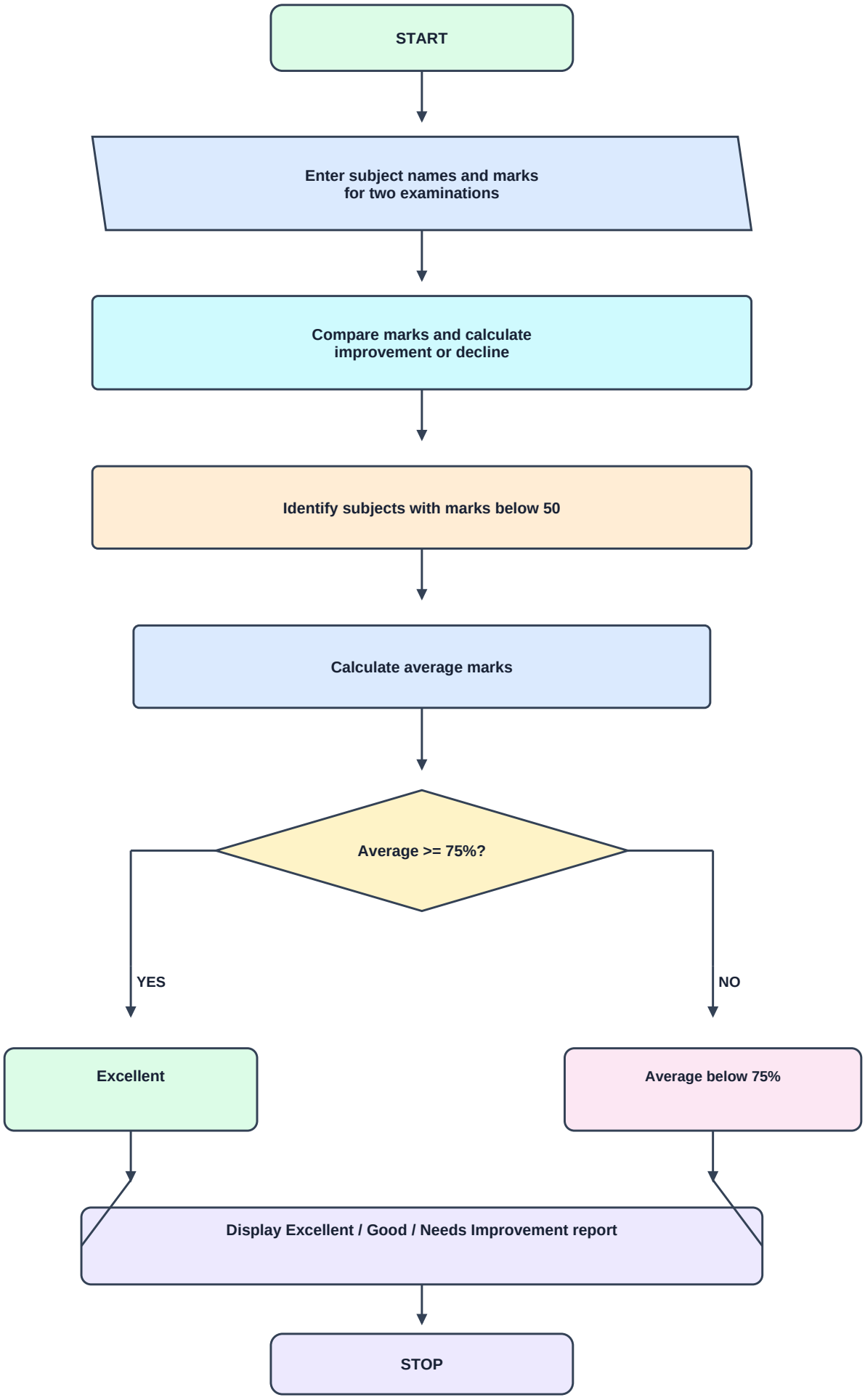
3. Assumptions

• Marks are between 0 and 100.
• Minimum passing marks are 50.
• Performance is classified based on average marks.

4. Algorithm

1. Start.
2. Enter subject names and examination marks.
3. Compare marks in two examinations.
4. Calculate improvement or decline.
5. Identify subjects with marks below 50.
6. Calculate average marks.
7. Classify student performance.
8. Display the results.
9. Stop.

5. Flowchart


START
Enter subject names and marks for two examinations
Compare marks and calculate improvement or decline
Identify subjects with marks below 50
Calculate average marks
Average >= 75%?
YES
NO
Excellent
Average below 75%
Display Excellent / Good / Needs Improvement report
STOP

6. Pseudocode

```
START
INPUT subject names and exam marks

FOR each subject
    change = exam2 - exam1
    DISPLAY change

    IF exam2 < 50 THEN
        DISPLAY "Weak Subject"
    END IF
END FOR

average = total marks / number of subjects

IF average >= 75 THEN
    DISPLAY "Excellent"
ELSE IF average >= 50 THEN
    DISPLAY "Good"
ELSE
    DISPLAY "Needs Improvement"
END IF

STOP
```

7. Python Program

```
subjects = ["Python", "Maths", "English"]

exam1 = [60, 45, 70]
exam2 = [75, 40, 80]

for i in range(len(subjects)):
    change = exam2[i] - exam1[i]

    print(subjects[i])
    print("Improvement/Decline:", change)

    if exam2[i] < 50:
        print("Weak Subject")
    else:
        print("Good Performance")

average = sum(exam2) / len(exam2)
print("Average Marks:", average)

if average >= 75:
    print("Excellent")
elif average >= 50:
    print("Good")
else:
    print("Needs Improvement")
```

8. Sample Output

```
Python
Improvement/Decline: 15
Good Performance

Maths
Improvement/Decline: -5
Weak Subject

English
Improvement/Decline: 10
Good Performance

Average Marks: 65.0
Good
```

9. Test Cases

Subject	Exam 1	Exam 2	Result
Python	60	75	Improved
Maths	45	40	Declined / Weak
English	70	80	Improved

10. Conclusion

This program compares examination marks, identifies improvements and weak subjects, and classifies student performance based on average marks.